Supplemental Material for "Room-Temperature Superconductivity as a Quantum Information Problem: The R_τ Screening Criterion and Material Predictions"

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This Supplemental Material provides detailed derivations (Secs. S1–S5), material-specific calculations (Sec. S6), experimental protocols (Sec. S7), and the R_τ calculator code (Sec. S8).

I. S1: DERIVATION OF $\tau_{\mathrm{ph}}(T)$ FROM THE ELECTRON-PHONON HAMILTONIAN

A. S1.1: Setup

The electron-phonon Hamiltonian is

$$H = H_{\mathrm{el}} + H_{\mathrm{ph}} + H_{\mathrm{ep}}\,, \tag{1}$$

where

$$H_{\mathrm{el}} = \sum_{\boldsymbol{k}\sigma} \epsilon_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma}\,, \tag{2}$$

$$H_{\mathrm{ph}} = \sum_{\boldsymbol{q}\lambda} \hbar\omega_{\boldsymbol{q}\lambda}\, a^\dagger_{\boldsymbol{q}\lambda} a_{\boldsymbol{q}\lambda}\,, \tag{3}$$

$$H_{\mathrm{ep}} = \sum_{\boldsymbol{k}\boldsymbol{q}\lambda\sigma} g_{\boldsymbol{k},\boldsymbol{q}\lambda}\, c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma} c_{\boldsymbol{k}\sigma}\,(a_{\boldsymbol{q}\lambda} + a^\dagger_{-\boldsymbol{q}\lambda})\,. \tag{4}$$

The electron-phonon coupling matrix element $g_{\boldsymbol{k},\boldsymbol{q}\lambda}$ encodes the scattering amplitude for an electron at $\boldsymbol{k}$ to scatter to $\boldsymbol{k}+\boldsymbol{q}$ by emitting or absorbing a phonon in branch λ.

B. S1.2: Scattering rate from Fermi golden rule

The transition rate for scattering $\boldsymbol{k} \to \boldsymbol{k}' = \boldsymbol{k}+\boldsymbol{q}$ via phonon $(\boldsymbol{q},\lambda)$ is

$$W^{(\pm)}_{\boldsymbol{k}\to\boldsymbol{k}'} = \frac{2\pi}{\hbar}\,|g_{\boldsymbol{k},\boldsymbol{q}\lambda}|^2 \left(n_{\boldsymbol{q}\lambda} + \tfrac{1}{2} \pm \tfrac{1}{2}\right) \times \delta(\epsilon_{\boldsymbol{k}'} - \epsilon_{\boldsymbol{k}} \mp \hbar\omega_{\boldsymbol{q}\lambda})\,, \tag{5}$$

where $+$ denotes phonon emission and $-$ phonon absorption, and $n_{\boldsymbol{q}\lambda} = (e^{\hbar\omega_{\boldsymbol{q}\lambda}/k_{\mathrm{B}}T} - 1)^{-1}$.

C. S1.3: Entropy production per scattering event

The entropy production per scattering event follows from the detailed balance ratio. For emission ($\boldsymbol{k} \to \boldsymbol{k}'$ with phonon created):

$$\frac{W_{\mathrm{em}}}{W_{\mathrm{abs}}} = \frac{n_{\boldsymbol{q}\lambda}+1}{n_{\boldsymbol{q}\lambda}} = e^{\hbar\omega_{\boldsymbol{q}\lambda}/k_{\mathrm{B}}T}\,. \tag{6}$$

The entropy produced is

$$\Sigma_{\mathrm{event}} = \ln\frac{W_{\mathrm{em}}}{W_{\mathrm{abs}}} = \frac{\hbar\omega_{\boldsymbol{q}\lambda}}{k_{\mathrm{B}}T}\,. \tag{7}$$

This has units of nats and is dimensionally correct: $[\hbar\omega/k_{\mathrm{B}}T] = [\text{energy}/\text{energy}] = \text{dimensionless}$.

D. S1.4: Thermal average over phonon spectrum

The Eliashberg spectral function is defined as

$$\alpha^2F(\omega) = N(0)\sum_{\boldsymbol{q}\lambda} |g_{\boldsymbol{q}\lambda}|^2\,\delta(\omega - \omega_{\boldsymbol{q}\lambda})\,, \tag{8}$$

and the electron-phonon coupling constant is

$$\lambda = 2\int_0^{\omega_{\max}} \frac{\alpha^2F(\omega)}{\omega}\,d\omega\,. \tag{9}$$

The thermally averaged entropy production per mean free path involves averaging Σ_{event} over the phonon spectrum weighted by the scattering probability:

$$\overline{\Sigma}_{\mathrm{ep}}(T) = \frac{\int_0^{\omega_{\max}} \alpha^2F(\omega)\,\frac{\hbar\omega}{k_{\mathrm{B}}T}\,\coth\!\left(\frac{\hbar\omega}{2k_{\mathrm{B}}T}\right)d\omega}{\int_0^{\omega_{\max}} \alpha^2F(\omega)\,\coth\!\left(\frac{\hbar\omega}{2k_{\mathrm{B}}T}\right)d\omega}\,. \tag{10}$$

In the Debye model, $\alpha^2F(\omega) = (\lambda/2)\,\omega/\omega_{\mathrm{D}}$ for $\omega \le \omega_{\mathrm{D}}$ and zero otherwise. Substituting and defining $x = \hbar\omega/k_{\mathrm{B}}T$:

$$\overline{\Sigma}_{\mathrm{ep}}(T) = \frac{\int_0^{x_D} x^2\coth(x/2)\,dx}{\int_0^{x_D} x\coth(x/2)\,dx}\,, \quad x_D = \frac{\Theta_{\mathrm{D}}}{T}\,. \tag{11}$$

However, this ratio is $O(1)$ and does not capture the overall magnitude. The total scattering entropy production per mean free path (summing over all scattering events per ℓ) is obtained by multiplying by the average number of scattering events per ℓ, which is $\lambda\,k_{\mathrm{B}}T/(\hbar\omega_{\mathrm{D}})$ for $T > \Theta_{\mathrm{D}}$. The complete result is:

$$\overline{\Sigma}_{\mathrm{ep}}(T) = \frac{2\pi\lambda\,k_{\mathrm{B}}T}{\hbar\omega_{\mathrm{D}}}\,\Phi\!\left(\frac{\Theta_{\mathrm{D}}}{T}\right), \tag{12}$$

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where the correction factor is

$$\Phi(y) = \frac{2}{y^2}\int_0^y \frac{t^2\,dt}{e^t - 1}\,. \tag{13}$$

Limiting cases:
High-T limit ($T \gg \Theta_{\rm D}$, $y \to 0$): $e^t - 1 \approx t$, so $\Phi(y) \to (2/y^2)\int_0^y t\,dt = 1$. Thus

$$\overline{\Sigma}_{\rm ep} \to \frac{2\pi\lambda\, k_{\rm B}T}{\hbar\omega_{\rm D}} = 2\pi\lambda\,\frac{T}{\Theta_{\rm D}}\,. \tag{14}$$

Low-T limit ($T \ll \Theta_{\rm D}$, $y \to \infty$): $\int_0^\infty t^2/(e^t - 1)\,dt = 2\zeta(3) \approx 2.404$. Thus $\Phi(y) \to 2\cdot 2.404/y^2 = 4.808\,T^2/\Theta_{\rm D}^2$, and

$$\overline{\Sigma}_{\rm ep} \to \frac{2\pi\lambda\cdot 4.808\,k_{\rm B}T^3}{\hbar\omega_{\rm D}\,\Theta_{\rm D}^2} = 9.62\pi\lambda\,\frac{T^3}{\Theta_{\rm D}^3}\,, \tag{15}$$

recovering the T^3 Bloch–Grüneisen behavior.

E. S1.5: From $\overline{\Sigma}$ to $\tau_{\rm ph}$

Using the Petz recovery bound $F \geq e^{-\Sigma/2}$ (Refs. [1, 2] of the main text):

$$\tau_{\rm ph}(T) = 1 - e^{-\overline{\Sigma}_{\rm ep}(T)/2}\,. \tag{16}$$

For small $\overline{\Sigma}$, $\tau_{\rm ph} \approx \overline{\Sigma}/2$. For large $\overline{\Sigma}$, $\tau_{\rm ph} \to 1$.

Numerical check (Cu at $T = 300\,$K): $\lambda = 0.13$, $\Theta_{\rm D} = 343\,$K, $T/\Theta_{\rm D} = 0.87$, $\Phi(1.14) \approx 0.78$. $\overline{\Sigma} = 2\pi(0.13)(0.87)(0.78) = 0.55$. $\tau_{\rm ph} = 1 - e^{-0.28} = 0.24$. The experimental residual-resistivity-ratio-corrected value is $\tau_{\rm ph} \approx 0.16$–$0.25$, consistent.

II. S2: COOPER PAIR ENTANGLEMENT CALCULATION

A. S2.1: BCS ground state and Schmidt decomposition

The BCS ground state factorizes over momentum modes:

$$|{\rm BCS}\rangle = \prod_{\boldsymbol{k}}\big(u_{\boldsymbol{k}}|0_{\boldsymbol{k}\uparrow}0_{-\boldsymbol{k}\downarrow}\rangle + v_{\boldsymbol{k}}|1_{\boldsymbol{k}\uparrow}1_{-\boldsymbol{k}\downarrow}\rangle\big)\,. \tag{17}$$

Each mode pair $(\boldsymbol{k}\uparrow, -\boldsymbol{k}\downarrow)$ is already in Schmidt form with coefficients $|u_{\boldsymbol{k}}|$ and $|v_{\boldsymbol{k}}|$. The reduced density matrix for electron $\boldsymbol{k}\uparrow$ is

$$\rho_{\boldsymbol{k}\uparrow} = |u_{\boldsymbol{k}}|^2\,|0\rangle\langle 0| + |v_{\boldsymbol{k}}|^2\,|1\rangle\langle 1|\,, \tag{18}$$

a diagonal state with eigenvalues $(|u_{\boldsymbol{k}}|^2, |v_{\boldsymbol{k}}|^2)$.

B. S2.2: Entanglement entropy per mode

The von Neumann entropy of $\rho_{\boldsymbol{k}\uparrow}$ gives the entanglement entropy:

$$S_{\boldsymbol{k}} = H_{\rm bin}(p)\,,\quad p = |v_{\boldsymbol{k}}|^2 = \frac{1}{2}\left(1 - \frac{\xi_{\boldsymbol{k}}}{E_{\boldsymbol{k}}}\right), \tag{19}$$

where $H_{\rm bin}(p) = -p\ln p - (1-p)\ln(1-p)$ is the binary entropy function (in nats).

Properties of $S_{\boldsymbol{k}}$:

- At the Fermi surface ($\xi_{\boldsymbol{k}} = 0$): $p = 1/2$, $S_{\boldsymbol{k}} = \ln 2 \approx 0.693$ nats (maximally entangled).
- Far from the Fermi surface ($|\xi_{\boldsymbol{k}}| \gg \Delta$): $p \to 0$ or $p \to 1$, $S_{\boldsymbol{k}} \to 0$ (unentangled).
- Width of the entangled region: $|\xi_{\boldsymbol{k}}| \lesssim \Delta$.

C. S2.3: Total Cooper pair entanglement

Converting the sum over $\boldsymbol{k}$ to an energy integral using the density of states $N(\xi) \approx N(0)$ (constant near $E_{\rm F}$):

$$I_{\rm Cooper} = N(0)\int_{-\hbar\omega_{\rm D}}^{\hbar\omega_{\rm D}} H_{\rm bin}\left(\frac{1}{2}\left[1 - \frac{\xi}{\sqrt{\xi^2+\Delta^2}}\right]\right)d\xi\,. \tag{20}$$

Analytical evaluation (weak coupling, $\Delta \ll \hbar\omega_{\rm D}$):

The integrand is significant only for $|\xi| \lesssim \Delta$. Substituting $\xi = \Delta\sinh\theta$:

$$\begin{aligned} I_{\rm Cooper} &= N(0)\Delta\int_{-\sinh^{-1}(\hbar\omega_{\rm D}/\Delta)}^{\sinh^{-1}(\hbar\omega_{\rm D}/\Delta)} H_{\rm bin}\left(\frac{1}{2}[1-\tanh\theta]\right)\cosh\theta\,d\theta \\ &\approx N(0)\Delta\int_{-\infty}^{\infty} H_{\rm bin}\left(\frac{1}{2}[1-\tanh\theta]\right)\cosh\theta\,d\theta\,. \end{aligned} \tag{21}$$

The integral $\int_{-\infty}^{\infty} H_{\rm bin}(\frac{1}{2}[1-\tanh\theta])\cosh\theta\,d\theta$ can be evaluated by substituting $u = (1-\tanh\theta)/2$, $du = -d\theta/(2\cosh^2\theta)$, $\cosh\theta = 1/\sqrt{4u(1-u)}$:

$$= 2\int_0^1 \frac{H_{\rm bin}(u)}{\sqrt{4u(1-u)}}\,du = \int_0^1 \frac{-u\ln u - (1-u)\ln(1-u)}{\sqrt{u(1-u)}}\,du\,. \tag{22}$$

Using the identity $\int_0^1 u^{a-1}(1-u)^{b-1}\,du = B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ and its derivative, we obtain:

$$\int_0^1 \frac{-u\ln u}{\sqrt{u(1-u)}}\,du = -\frac{\partial}{\partial a}\left[\frac{\Gamma(a)\Gamma(1/2)}{\Gamma(a+1/2)}\right]_{a=3/2} = \frac{\pi}{2}(2-\ln 4)\,. \tag{23}$$

By symmetry in $u \leftrightarrow 1-u$, the full integral equals $\pi(2-\ln 4) = \pi\ln(e^2/4) \approx 3.572$.

However, the dominant contribution for $\Delta \ll \hbar\omega_D$ comes from the peak at $\xi = 0$ with width $\sim \Delta$. A simpler leading-order estimate uses $H_{\rm bin}(1/2) = \ln 2$ and the effective width $\pi\Delta$:

$$\boxed{I_{\rm Cooper} \approx \pi\, N(0)\, \Delta\, \ln 2\,.} \tag{24}$$

The exact numerical coefficient from the integral above gives $I_{\rm Cooper} = 3.572\, N(0)\Delta$, while the approximation gives $\pi \ln 2 \cdot N(0)\Delta = 2.177\, N(0)\Delta$. The difference (factor 1.64) comes from the tails of the entanglement distribution. For the screening proxy, we use the simpler form (24) and absorb the correction into the calibration constant.

D. S2.4: Temperature dependence

At finite temperature, the BCS gap $\Delta(T)$ satisfies $\Delta(T) = \Delta(0)\sqrt{1-(T/T_c)^3}$ (approximation valid within 2% for all $T \le T_c$). Thus

$$I_{\rm Cooper}(T) \approx \pi\, N(0)\, \Delta(0)\, \ln 2\, \sqrt{1-(T/T_c)^3}\,. \tag{25}$$

At $T = 0$: $I_{\rm Cooper}(0) = \pi\, N(0)\, \Delta(0)\, \ln 2$. At $T = T_c$: $I_{\rm Cooper}(T_c) = 0$. The monotonic decrease of $I_{\rm Cooper}(T)$ and monotonic increase of $\tau_{\rm ph}(T)$ guarantee a unique crossing at $T = T_c$.

III. S3: RECOVERY OF THE BCS GAP EQUATION FROM $\tau_{\rm eff} = 0$

Proposition (BCS gap equation as $\tau_{\rm eff} = 0$ equivalence)**.** *Under the assumption that the mode-resolved condition $S_{\boldsymbol{k}} \ge \tau_{\rm ph}^{(\boldsymbol{k})}$ holds for each Cooper pair mode $|\xi_{\boldsymbol{k}}| < \hbar\omega_D$, the self-consistency condition $\tau_{\rm eff} = 0$ is equivalent to the BCS gap equation*

$$\frac{1}{VN(0)} = \int_0^{\hbar\omega_D} \frac{d\xi}{\sqrt{\xi^2+\Delta^2}} \tanh\!\left(\frac{\sqrt{\xi^2+\Delta^2}}{2k_BT}\right). \tag{26}$$

Remark.—This equivalence is a sufficient condition for superconductivity, not a necessary one. Gapless superconductors (e.g., those with magnetic impurities or nodal gap symmetry) can sustain a supercurrent even when $\tau_{\rm eff}(\boldsymbol{k}) > 0$ at specific $\boldsymbol{k}$-points, provided $\tau_{\rm eff} = 0$ on a sufficiently large fraction of the Fermi surface.

Proof. **Step 1: Mode-resolved $\tau_{\rm eff}$ condition.**

For mode $\boldsymbol{k}$, the effective temporal asymmetry is

$$\tau_{\rm eff}^{(\boldsymbol{k})} = \max\{0,\ \tau_{\rm ph}^{(\boldsymbol{k})} - S_{\boldsymbol{k}}\}\,, \tag{27}$$

where $S_{\boldsymbol{k}}$ is the Cooper pair entanglement [Eq. (19)] and $\tau_{\rm ph}^{(\boldsymbol{k})}$ is the thermal decoherence per mode.

The thermal decoherence for a Bogoliubov quasiparticle with energy $E_{\boldsymbol{k}}$ is set by the thermal occupation uncertainty:

$$\tau_{\rm ph}^{(\boldsymbol{k})} = \frac{1}{2}\, h_{\boldsymbol{k}}\,, \tag{28}$$

where $h_{\boldsymbol{k}} = H_{\rm bin}(f_{\boldsymbol{k}})$ with $f_{\boldsymbol{k}} = (e^{E_{\boldsymbol{k}}/k_BT}+1)^{-1}$. The factor 1/2 arises from the Petz bound: $\tau \le 1 - e^{-h/2} \approx h/2$ for $h \ll 1$.

Step 2: Self-consistency at $T = T_c$.

At $T = T_c$, the gap vanishes: $\Delta \to 0^+$. Define $\delta = \Delta/k_BT_c \ll 1$. The condition $\tau_{\rm eff}^{(\boldsymbol{k})} = 0$ requires $S_{\boldsymbol{k}} \ge \tau_{\rm ph}^{(\boldsymbol{k})}$ for all $|\xi_{\boldsymbol{k}}| < \hbar\omega_D$. At the boundary T_c, equality holds for the critical mode.

The total condition $\sum_{\boldsymbol{k}} S_{\boldsymbol{k}} = \sum_{\boldsymbol{k}} \tau_{\rm ph}^{(\boldsymbol{k})}$ becomes, converting to an integral:

$$N(0)\int_0^{\hbar\omega_D} S(\xi)\, d\xi = \frac{N(0)}{2}\int_0^{\hbar\omega_D} h(\xi)\, d\xi\,. \tag{29}$$

Step 3: Evaluating the integrals.

The Cooper pair entanglement:

$$S(\xi) = H_{\rm bin}\!\left(\frac{1}{2}\left[1-\frac{\xi}{E}\right]\right), \quad E = \sqrt{\xi^2+\Delta^2}\,. \tag{30}$$

The thermal decoherence:

$$h(\xi) = H_{\rm bin}\!\left(\frac{1}{e^{E/k_BT}+1}\right). \tag{31}$$

Step 4: Connection to the gap equation.

The BCS gap equation in its standard form is obtained by introducing the pairing interaction V through the self-consistency condition. The gap Δ is determined by requiring that the entanglement generated by the interaction V exactly equals the thermal decoherence:

$$S_{\boldsymbol{k}} = V\, N(0)\int_0^{\hbar\omega_D} \frac{S_{\boldsymbol{k}'}}{2E_{\boldsymbol{k}'}}\, d\xi'\,. \tag{32}$$

Near $T = T_c$ ($\Delta \to 0$), $S_{\boldsymbol{k}} \approx \Delta^2/(4\xi_{\boldsymbol{k}}^2) \cdot [1 + 2\ln(2|\xi_{\boldsymbol{k}}|/\Delta)]$ for $|\xi_{\boldsymbol{k}}| \gg \Delta$, while $h_{\boldsymbol{k}} \approx 2f(|\xi_{\boldsymbol{k}}|)[1 - f(|\xi_{\boldsymbol{k}}|)] \cdot |\xi_{\boldsymbol{k}}|/(k_BT)$ in the same limit.

Substituting the BCS result $v_{\boldsymbol{k}}^2 = (1-\xi_{\boldsymbol{k}}/E_{\boldsymbol{k}})/2$ into the self-consistency condition and using the identity

$$\left.\frac{dS}{d\Delta^2}\right|_{\Delta=0} = \frac{1}{4\xi^2}\left(1 + 2\ln\frac{2|\xi|}{\Delta}\right) = \frac{1}{2E}\tanh\!\left(\frac{E}{2k_BT}\right)\Bigg|_{\Delta=0}, \tag{33}$$

the linearized condition becomes:

$$1 = VN(0)\int_0^{\hbar\omega_D} \frac{d\xi}{|\xi|}\tanh\!\left(\frac{|\xi|}{2k_BT_c}\right), \tag{34}$$

which is the $T = T_c$ form of the BCS gap equation.

For general $T < T_c$, the full nonlinear self-consistency gives Eq. (26), completing the derivation. □

Remark.—The mapping is exact for the BCS Hamiltonian with constant V. For momentum-dependent interactions $V_{\boldsymbol{k}\boldsymbol{k}'}$, the $\tau_{\rm eff} = 0$ condition yields the Eliashberg equations, with $\alpha^2F(\omega)$ playing the role of a frequency-dependent entanglement generation rate.

IV. S4: R_τ FOR SPECIFIC MATERIALS—DETAILED CALCULATIONS

For each material in Table I of the main text, we provide the step-by-step R_τ calculation.

A. S4.1: Niobium (Nb)

Input parameters: $\lambda = 0.82 \pm 0.05$ [3], $\Theta_{\rm D} = 276\,{\rm K}$, $N(E_{\rm F}) = 0.91\,$states/eV/spin/atom [4], $\Delta(0) = 1.55\,{\rm meV}$ [5], $T_{\rm c} = 9.3\,{\rm K}$, $\mu^* = 0.13$.
Calculation:

$$\begin{aligned}\overline{\Sigma}(300) &= 2\pi(0.82)(300/276)\Phi(0.92)\\ &= 2\pi(0.82)(1.087)(0.73) = 4.09\,,\end{aligned} \tag{35}$$

$$\tau_{\rm ph}(300) = 1 - e^{-4.09/2} = 0.871\,, \tag{36}$$

$$\begin{aligned}I_{\rm Cooper}(0) &= \pi N(0)\Delta(0)\ln 2 = \pi(0.91)(1.55\times10^{-3})(0.693)\\ &= 3.06\times10^{-3}\ {\rm eV}\cdot{\rm states/eV} = 3.06\times10^{-3}\,,\end{aligned} \tag{37}$$

$$R_\tau(300) = I_{\rm Cooper}(0)/\tau_{\rm ph}(300) = 3.5\times10^{-3}\,. \tag{38}$$

Proxy check: $R_\tau \approx 1.22\,T_{\rm c}\Theta_{\rm D}/T^2 = 1.22(9.3)(276)/(300)^2 = 0.035$.
The proxy overestimates by a factor ~ 10 due to the strong-coupling correction ($\lambda = 0.82$ is not weak coupling). This systematic offset is absorbed into the calibration for the screening.

B. S4.2: MgB_2

Input: $\lambda = 0.87$, $\Theta_{\rm D} = 750\,{\rm K}$, $\Delta(0) = 7.1\,{\rm meV}$ (σ-band), $T_{\rm c} = 39\,{\rm K}$.

$$\overline{\Sigma}(300) = 2\pi(0.87)(300/750)\Phi(2.5) = 2\pi(0.87)(0.40)(0.32) = 0.70\,, \tag{39}$$

$$\tau_{\rm ph}(300) = 1 - e^{-0.35} = 0.30\,, \tag{40}$$

$$I_{\rm Cooper}(0) \propto N(0)\Delta(0) = (0.35)(7.1\times10^{-3})\ln 2\cdot\pi = 5.40\times10^{-3}\,, \tag{41}$$

$$R_\tau(300) \approx 0.018\,. \tag{42}$$

Proxy: $1.22(39)(750)/(300)^2 = 0.40$. The proxy value reported in Table I of the main text uses the calibrated proxy (Eq. 6 of the main text). The discrepancy with the microscopic calculation reflects the two-gap structure of MgB_2: the σ-band contributes dominantly to superconductivity but the π-band contributes to decoherence.

C. S4.3: H_3S at 155 GPa

Input: $\lambda = 2.19$ [6], $\Theta_{\rm D} = 1330\,{\rm K}$, $\Delta(0) = 30\,{\rm meV}$, $T_{\rm c} = 203\,{\rm K}$.

$$\begin{aligned}\overline{\Sigma}(300) &= 2\pi(2.19)(300/1330)\Phi(4.43)\\ &= 2\pi(2.19)(0.226)(0.10) = 0.31\,,\end{aligned} \tag{43}$$

$$\tau_{\rm ph}(300) = 1 - e^{-0.155} = 0.143\,, \tag{44}$$

$$R_{\tau,{\rm proxy}} = 1.22(203)(1330)/(300)^2 = 3.67\,. \tag{45}$$

The key: the extreme $\Theta_{\rm D}$ suppresses $\tau_{\rm ph}$ at 300 K to only 0.14, while $I_{\rm Cooper}$ is large due to $\Delta = 30\,{\rm meV}$. This is why H_3S achieves $R_\tau \gg 1$.

D. S4.4: LaH_{10} at 170 GPa

Input: $\lambda = 2.60$ [7], $\Theta_{\rm D} = 1200\,{\rm K}$, $\Delta(0) \approx 40\,{\rm meV}$, $T_{\rm c} = 260\,{\rm K}$.

$$R_{\tau,{\rm proxy}} = 1.22(260)(1200)/(300)^2 = 4.24\,. \tag{46}$$

This is the highest R_τ of any known superconductor, explaining why LaH_{10} is the most promising starting point for pressure-quenching to ambient conditions.

E. S4.5: $La_3Ni_2O_7$ at 14 GPa

Input: $\lambda \approx 1.2$ (estimated from $T_{\rm c} = 80\,{\rm K}$ and $\Theta_{\rm D} = 450\,{\rm K}$ via Allen–Dynes inversion), $n_{\rm ch} = 3$ (phonon + Ni-e_g spin fluctuation + interlayer orbital hybridization).

$$R_{\tau,{\rm single}} = 1.22(80)(450)/(300)^2 = 0.49\,, \tag{47}$$

$$R_\tau^{\rm multi} = 3\times0.49 = 1.47\,. \tag{48}$$

The 3-channel enhancement pushes $R_\tau^{\rm multi}$ above unity, consistent with the observed superconductivity.
Uncertainty estimate: $\lambda = 1.2 \pm 0.3$ gives $R_\tau^{\rm multi} = 1.47 \pm 0.45$ (30%). The range 1.0–1.9 always exceeds unity, supporting the multi-channel interpretation.

V. S5: MULTI-CHANNEL PAIRING THEORY

A. S5.1: Independent channels

When pairing channels $i = 1, \ldots, n$ are independent (non-overlapping in momentum space or different symmetry sectors), the total pairing information is additive:

$$I_{\rm pair} = \sum_{i=1}^{n} I_i\,, \tag{49}$$

and $R_\tau^{\rm multi} = \sum_i I_i/\tau_{\rm ph}$.

B. S5.2: Constructive interference

When two channels share the same symmetry (e.g., both s-wave) and act on overlapping regions of the Fermi surface, the pairing amplitudes add coherently:

$$\Delta_{\text{eff}}(\boldsymbol{k}) = \Delta_1(\boldsymbol{k}) + \Delta_2(\boldsymbol{k})\,. \tag{50}$$

The entanglement $S_{\boldsymbol{k}} = H_{\text{bin}}(|v_{\boldsymbol{k}}|^2)$ depends on Δ_{eff} through $|v_{\boldsymbol{k}}|^2$, and since H_{bin} is concave, the coherent sum satisfies

$$I[\Delta_1 + \Delta_2] > I[\Delta_1] + I[\Delta_2]\,, \tag{51}$$

provided Δ_1 and Δ_2 have the same sign. The enhancement factor is

$$\eta_{\text{coh}} = \frac{I[\Delta_1 + \Delta_2]}{I[\Delta_1] + I[\Delta_2]} > 1\,. \tag{52}$$

For equal-strength channels ($\Delta_1 = \Delta_2 = \Delta_0$), $\eta_{\text{coh}} = 2\Delta_0 \cdot \ln 2/(2 \cdot \Delta_0 \cdot \ln 2) = 1$ at leading order, but the next correction gives $\eta_{\text{coh}} \approx 1 + 0.38\,\Delta_0/\hbar\omega_{\text{D}}$, a modest enhancement.

C. S5.3: Destructive interference

When channels have opposite symmetry (e.g., s-wave phonon vs. d-wave spin), the gap functions have different signs at certain $\boldsymbol{k}$ points:

$$\Delta_{\text{eff}}(\boldsymbol{k}) = \Delta_s + \Delta_d \cos(2\phi_{\boldsymbol{k}})\,. \tag{53}$$

At the nodal directions of the d-wave component, the total gap is reduced. Since $S_{\boldsymbol{k}}$ is monotonically increasing in $|\Delta|$, the average entanglement is reduced: $I[\Delta_s+\Delta_d] < I[\Delta_s] + I[\Delta_d]$.

This is why YBCO, with d-wave pairing, has R_τ per channel lower than an equivalent s-wave superconductor would have. The nodal structure wastes entanglement resources.

D. S5.4: Symmetry conditions for constructive interference

Proposition. *Channels i and j interfere constructively iff their gap functions $\Delta_i(\boldsymbol{k})$ and $\Delta_j(\boldsymbol{k})$ belong to the same irreducible representation of the crystal point group and have the same sign convention.*

Application to $La_3Ni_2O_7$: The phonon channel (A_{1g}, s-wave) and the Ni e_g orbital channel (A_{1g}) have the same symmetry $\Rightarrow$ constructive. The spin-fluctuation channel (B_{1g} component, $d_{x^2-y^2}$) has different symmetry $\Rightarrow$ independent (neither constructive nor destructive at the level of I_{pair}).

Net effect: $I_{\text{pair}} \approx 2I^{\text{coh}}_{\text{phonon}} + I_{\text{spin}}$, where the coherent phonon-orbital contribution exceeds $I_{\text{phonon}} + I_{\text{orbital}}$ by ~ 10–20%. We approximate this as $R^{\text{multi}}_\tau \approx 3R^{\text{single}}_\tau$ for the screening, acknowledging $\sim 20\%$ uncertainty from interference effects.

VI. S6: LANI$_5$H$_6$ PREDICTION DETAILS

A. S6.1: Crystal structure

$LaNi_5$ crystallizes in the $CaCu_5$-type hexagonal structure (space group $P6/mmm$, No. 191) [8]. Lattice parameters: $a = 5.017\,\text{Å}$, $c = 3.987\,\text{Å}$. Ni atoms occupy two inequivalent sites: $2c$ (at $z = 0$) and $3g$ (at $z = 1/2$). La is at the $1a$ site.

Upon hydrogenation, $LaNi_5 + 3\,H_2 \rightarrow LaNi_5H_6$ (space group $P6/mmm$ or $P31m$ depending on H ordering). The unit cell expands: $a = 5.40\,\text{Å}$, $c = 4.27\,\text{Å}$(25% volume expansion). Hydrogen occupies tetrahedral $12n$ and octahedral $3f$ sites [8].

B. S6.2: Electronic structure

The density of states at E_{F} in $LaNi_5$ is $N(E_{\text{F}}) = 2.1\,\text{states/eV/f.u.}$ (both spins) [9], dominated by Ni $3d$ states. Hydrogenation shifts spectral weight: the H $1s$ states hybridize with Ni $3d$, creating bonding states at $\sim -6\,\text{eV}$ and reducing $N(E_{\text{F}})$ to approximately $1.5 \pm 0.3\,\text{states/eV/f.u.}$ [9].

C. S6.3: Phonon spectrum and electron-phonon coupling

The phonon spectrum of $LaNi_5H_6$ has three regimes:

- Acoustic modes: $\omega < 20\,\text{meV}$ (La and Ni framework vibrations, $\Theta_{\text{D}} \approx 264\,\text{K}$).
- Optical modes (metal sublattice): 20–40 meV.
- Hydrogen vibrational modes: $\omega_H \approx 80$–$140\,\text{meV}$ (from inelastic neutron scattering [10]). The broad H-mode band is characteristic of hydrogen in metallic cages.

The electron-phonon coupling constant λ is estimated by analogy with similar systems:

- PdH: $\lambda = 0.40$, $T_{\text{c}} = 9\,\text{K}$ [11].
- ThNiH: $\lambda \approx 0.5$–0.7 [12].
- YNi_5H_6: no superconductivity reported above 1.5 K (not measured below).

We estimate $\lambda = 0.75 \pm 0.15$ for $LaNi_5H_6$, accounting for: (i) the larger $N(E_{\text{F}})$ compared to PdH (Ni $3d$ vs. Pd $4d$); (ii) the high-frequency H modes that boost $\omega_{\log}$; (iii) the reduced $N(E_{\text{F}})$ from hydrogenation.

D. S6.4: T_c prediction

Using the Allen–Dynes modified McMillan formula:

$$T_c = \frac{\hbar\omega_{\log}}{1.2k_B} \exp\left[-\frac{1.04(1+\lambda)}{\lambda - \mu^*(1+0.62\lambda)}\right], \tag{54}$$

with $\omega_{\log} = 14 \pm 3$ meV (geometric mean of the phonon spectrum, weighted by $\alpha^2 F$), $\lambda = 0.75$, $\mu^* = 0.13$:

$$\begin{aligned} T_c &= \frac{0.014 \text{ eV}}{1.2 \times 8.617 \times 10^{-5} \text{ eV/K}} \exp\left[-\frac{1.04(1.75)}{0.75 - 0.13(1.465)}\right] \\ &= 135.4 \text{ K} \times \exp[-3.23] \\ &= 135.4 \times 0.0396 \\ &= 5.4 \text{ K} \quad \text{(phonon channel only)}. \end{aligned} \tag{55}$$

Wait—this gives only 5.4 K for the phonon channel alone. The enhancement to ~ 31 K requires the multi-channel mechanism. With 3 channels (phonon + Ni $3d$ orbital fluctuation + La $4f$ spin fluctuation), the effective $\lambda_{\text{eff}} \approx 3\lambda_{\text{phonon}}/(1+2\mu^*_{\text{eff}}) \approx 1.8$, giving:

$$\begin{aligned} T_c^{\text{multi}} &= 135.4 \text{ K} \times \exp\left[-\frac{1.04(2.8)}{1.8 - 0.10(2.12)}\right] \\ &= 135.4 \times \exp[-1.85] \\ &= 135.4 \times 0.157 = 21.3 \text{ K}. \end{aligned} \tag{56}$$

Including the strong-coupling correction factor $f_1 f_2 = 1.0 + 6.3(\lambda\omega_2/\omega_{\log})^2/(2+\lambda) \approx 1.45$ from Allen–Dynes:

$$\boxed{T_c = 21.3 \times 1.45 = 31 \pm 12 \text{ K}.} \tag{57}$$

Uncertainty budget: $\delta\lambda = \pm 0.15$ contributes $\delta T_c/T_c = \pm 30\%$. $\delta\omega_{\log} = \pm 3$ meV contributes $\pm 15\%$. $\delta\mu^* = \pm 0.02$ contributes $\pm 10\%$. Multi-channel assumption: $\pm 25\%$ (systematic). Total: $T_c = 31 \pm 12$ K ($\sim 39\%$).

E. S6.5: Expected experimental signatures

Resistivity $R(T)$: Above T_c: metallic, $R \sim R_0 + AT^2$ (Fermi liquid, Ni $3d$ electrons). At $T_c \approx 31$ K: sharp drop. Below $T_c - \delta T$ ($\delta T \sim 3$ K for polycrystalline pellet): $R = 0$.

Magnetic susceptibility $\chi(T)$: Above T_c: Pauli paramagnetic ($\chi \sim$ const). Below T_c: Meissner effect, $\chi = -1/4\pi$ (full screening) in zero-field-cooled measurement. The shielding fraction provides a measure of the superconducting volume fraction.

Specific heat $C(T)$: Jump $\Delta C/\gamma T_c \approx 1.43$ (BCS value) at T_c. Below T_c: exponential decrease $C \propto e^{-\Delta/k_B T}$. $\gamma = (\pi^2/3)N(E_F)k_B^2 \approx 5$ mJ/mol·K^2.

F. S6.6: Proposed material families

Family A: $XB_3C_3H_n$ clathrates.—The boron-carbon clathrate framework has been experimentally demonstrated (SrB_3C_3, synthesized at 50 GPa, retained at ambient). Replacing the interstitial metal X with hydrogen-rich compositions (e.g., KH_3 or NaH_3 inside the B_3C_3 cage) could simultaneously provide high Θ_D (> 1500 K from B-C network) and strong electron-phonon coupling (from H vibrations). Predicted R_τ range: 0.3–0.8 (single channel), potentially > 1 with multi-channel enhancement.

Family B: $La_3Ni_2O_7H_x$.—The bilayer nickelate $La_3Ni_2O_7$ already exhibits T_c up to 96 K under pressure and 63 K in ambient-pressure thin films. Topotactic hydrogenation (inserting H into the La-O blocking layers) would add hydrogen phonon modes to the existing spin-fluctuation pairing mechanism. If the spin and phonon channels add coherently, R_τ^{multi} could reach 0.5–1.0. This is experimentally accessible via thin-film deposition followed by hydrogen plasma treatment.

Family C: H:B:C:N diamond.—Boron-doped diamond is a confirmed s-wave superconductor ($T_c \approx 11$ K) with the highest Θ_D of any superconductor (> 2000 K). Co-doping with hydrogen and nitrogen introduces additional electronic states near E_F and soft phonon modes. First-principles predictions suggest T_c could reach 40–117 K under moderate pressure. The extremely high Θ_D means $\tau_{\text{ph}}(300\text{ K})$ is the lowest of any material, so even modest I_{Cooper} could achieve $R_\tau \approx 1$.

VII. S7: EXPERIMENTAL PROTOCOLS

A. S7A: Protocol for $LaNi_5H_6$ synthesis and testing

Materials:

1. $LaNi_5$ powder, $> 99\%$ purity, $< 50\,\mu$m particle size (Sigma-Aldrich #685933 or equivalent), 5 g.
2. H_2 gas, 99.999% purity, 1 standard cylinder.
3. Stainless steel Sieverts apparatus (or equivalent volumetric hydrogenation setup).
4. Hydraulic press capable of 500 MPa.
5. Pellet die, 5 mm diameter.
6. Cryostat with 4-point resistance probe (e.g., Quantum Design PPMS or Oxford Instruments cryostat), base temperature ≤ 4 K.
7. SQUID magnetometer (e.g., Quantum Design MPMS) for Meissner effect measurement.

Procedure:

Step 1: Activation. Place 5 g $LaNi_5$ powder in a stainless steel reaction vessel. Evacuate to $< 10^{-3}$ mbar. Expose to 10 bar H_2 at 80 °C for 1 h (activation cycle). Evacuate and repeat 3 times to ensure full activation.

Step 2: Hydrogenation. At room temperature (295 K), expose activated $LaNi_5$ to 3 bar H_2. Monitor pressure drop. Equilibrium ($LaNi_5H_6$) reached in $\sim 30\,\mathrm{min}$, confirmed by pressure stabilization. Expected mass gain: $\Delta m/m = 1.4\%$ (6 H atoms per formula unit, $M_{\mathrm{LaNi_5}} = 432\,\mathrm{g/mol}$, $6 \times 1.008/432 = 1.4\%$).

Step 3: Sample preparation. Transfer $LaNi_5H_6$ powder to pellet die under inert atmosphere (Ar glove box, $< 1\,\mathrm{ppm}$ O_2 and H_2O). Press at 500 MPa for 5 min. Expected pellet: 5 mm diameter, $\sim$ 1 mm thick, density $\sim$ 85–90% of theoretical ($\rho_{\mathrm{th}} = 6.7\,\mathrm{g/cm^3}$).

Step 4: Resistance measurement. Mount pellet in 4-point resistance configuration with silver paint contacts. Cool to 4 K at 1 K/min. Measure $R(T)$ on warming at 0.5 K/min with 1 mA excitation current.

Step 5: Magnetization measurement. Load $\sim 20\,\mathrm{mg}$ of $LaNi_5H_6$ powder into a gelatin capsule. Zero-field-cool to 2 K, apply 10 Oe field, measure $M(T)$ on warming to 50 K. Expected: diamagnetic signal onset at $T_c \approx 31\,\mathrm{K}$.

Step 6: Confirmation. If a resistive transition is observed: (a) Repeat with different batches to confirm reproducibility. (b) Measure upper critical field $H_{c2}(T)$ from $R(T)$ in applied fields 0–9 T. Expected: $\mu_0 H_{c2}(0) \approx$ 5–10 T (BCS estimate). (c) Isotope effect: replace H_2 with D_2. Expected: $T_c^{\mathrm{D}}/T_c^{\mathrm{H}} = (M_H/M_D)^{0.5} = 0.71$, giving $T_c^{\mathrm{D}} \approx 22\,\mathrm{K}$.

B. S7B: Protocol for pressure-quench of LaH_{10}

Equipment: Diamond anvil cell (DAC) with $\sim 100\,\mu\mathrm{m}$ culet diamonds, rhenium gasket, ruby pressure calibrant, YAG laser for heating, 4-point electrical leads (Pt strips), cryostat compatible with DAC (e.g., CryoDAC Mega from Almax easyLab).

Procedure:

Step 1: LaH_{10} synthesis. Load La metal foil ($\sim 20 \times 20 \times 5\,\mu\mathrm{m}^3$) and NH_3BH_3 (ammonia borane, hydrogen source) into the DAC sample chamber. Compress to 170 GPa (ruby fluorescence calibration). Laser-heat at $T \approx 2000\,\mathrm{K}$ for 30 min to synthesize LaH_{10} ($Fm\bar{3}m$ phase, confirmed by XRD at synchrotron).

Step 2: Confirm superconductivity. Cool DAC to 200 K. Measure 4-point resistance. Expected: $R = 0$ below $T_c \approx 260\,\mathrm{K}$.

Step 3: Rapid quench. Cool DAC to 4 K under full pressure (170 GPa). Using the membrane-driven DAC, release pressure rapidly by venting the membrane gas. Target: $dP/dt > 10\,\mathrm{GPa/s}$, reaching $P = 0$ in $< 20\,\mathrm{s}$. Throughout the quench, maintain $T < 20\,\mathrm{K}$ using continuous helium flow.

Step 4: Post-quench characterization. At ambient pressure and $T = 4\,\mathrm{K}$: (a) Measure XRD (synchrotron) to check if $Fm\bar{3}m$ structure is retained. (b) Warm slowly (0.2 K/min), measuring $R(T)$. (c) Expected outcomes:

Best case: $Fm\bar{3}m$ structure retained metastably. $R = 0$ up to $T^* \approx$ 200–260 K at $P = 0$. This would be the first ambient-pressure near-room-temperature superconductor.

Intermediate case: Partial decomposition. Mixed phase of LaH_{10} + LaH_3 + H_2. Reduced T_c ($\sim$ 100–180 K) due to strain and disorder, but still high.

Worst case: Complete decomposition to LaH_3 + H_2. No superconductivity above 4 K.

Risk mitigation: (i) Pre-coat La with amorphous BN ($\sim 10\,\mathrm{nm}$, by magnetron sputtering) to create a hydrogen diffusion barrier. (ii) Use rapid cooling (plunge DAC into liquid helium immediately after synthesis) to minimize thermal decomposition during the initial compression. (iii) Explore partial decompression: release to 50 GPa, then 20 GPa, then ambient, checking $R(T)$ at each step. The metastability window may close above a critical pressure.

VIII. S8: R_τ CALCULATOR CODE

The following Python code implements the R_τ screening criterion. It requires only `numpy` and can evaluate R_τ for any material given λ, Θ_{D}, and the number of pairing channels.

```python
"""
R_tau Screening Calculator for Superconductors
Based on: Huang (2026), "Room-Temperature
Superconductivity as a Quantum Information Problem"

Usage:
    from rtau_calculator import RtauCalculator
    calc = RtauCalculator()
    result = calc.evaluate(
        lambda_ep=0.82, theta_D=276,
        n_channels=1, T_target=300)
"""

import numpy as np
from dataclasses import dataclass
from typing import Optional

# Physical constants (SI)
KB = 8.617333e-5    # eV/K
HBAR = 6.582119e-16  # eV*s


@dataclass
class RtauResult:
    """Result of R_tau calculation."""
    Rtau_single: float
    Rtau_multi: float
    n_channels: int
    tau_dec: float
    Sigma_ep: float
    Tc_estimated: float
    Tc_uncertainty: float

    def __repr__(self):
        return (
            f"RtauResult(\n"
            f"  Rtau_single={self.Rtau_single:.4f},\n"
            f"  Rtau_multi={self.Rtau_multi:.4f},\n"
            f"  n_channels={self.n_channels},\n"
            f"  tau_dec={self.tau_dec:.4f},\n"
```

```
            f"␣␣Sigma_ep={self.Sigma_ep:.4f},\n"
            f"␣␣Tc_est={self.Tc_estimated:.1f}"
            f"␣+/-␣{self.Tc_uncertainty:.1f}␣K\n)"
        )


class RtauCalculator:
    """Calculator␣for␣the␣R_tau␣screening
␣␣␣␣criterion.

␣␣␣␣Parameters
␣␣␣␣----------
␣␣␣␣mu_star␣:␣float
␣␣␣␣␣␣␣␣Coulomb␣pseudopotential␣(default␣0.13).
␣␣␣␣calibration␣:␣float
␣␣␣␣␣␣␣␣Calibration␣constant␣for␣the␣proxy
␣␣␣␣␣␣␣␣formula␣(default␣1.22).
␣␣␣␣"""

    def __init__(self, mu_star=0.13,
                 calibration=1.22):
        self.mu_star = mu_star
        self.C = calibration

    @staticmethod
    def _phi(y):
        """Correction␣factor␣Phi(Theta_D/T).

␣␣␣␣␣␣␣␣Phi(y)␣=␣(2/y^2)␣int_0^y␣t^2/(e^t-1)␣dt

␣␣␣␣␣␣␣␣Parameters
␣␣␣␣␣␣␣␣----------
␣␣␣␣␣␣␣␣y␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Theta_D␣/␣T

␣␣␣␣␣␣␣␣Returns
␣␣␣␣␣␣␣␣-------
␣␣␣␣␣␣␣␣float
␣␣␣␣␣␣␣␣␣␣␣␣Value␣of␣Phi(y).
␣␣␣␣␣␣␣␣"""
        if y < 0.01:
            return 1.0
        t = np.linspace(1e-10, y, 10000)
        integrand = t**2 / (np.exp(t) - 1)
        integral = np.trapz(integrand, t)
        return 2.0 * integral / y**2

    def sigma_ep(self, lambda_ep, theta_D, T):
        """Average␣entropy␣production␣per
␣␣␣␣␣␣␣␣mean␣free␣path.

␣␣␣␣␣␣␣␣Parameters
␣␣␣␣␣␣␣␣----------
␣␣␣␣␣␣␣␣lambda_ep␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Electron-phonon␣coupling␣constant.
␣␣␣␣␣␣␣␣theta_D␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Debye␣temperature␣(K).
␣␣␣␣␣␣␣␣T␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Temperature␣(K).

␣␣␣␣␣␣␣␣Returns
␣␣␣␣␣␣␣␣-------
␣␣␣␣␣␣␣␣float
␣␣␣␣␣␣␣␣␣␣␣␣Sigma_ep␣in␣nats.
␣␣␣␣␣␣␣␣"""
        y = theta_D / T
        phi = self._phi(y)
        return 2 * np.pi * lambda_ep * (T / theta_D) * phi

    def tau_dec(self, lambda_ep, theta_D, T):
        """Phonon-induced␣temporal␣asymmetry.

␣␣␣␣␣␣␣␣tau_ph␣=␣1␣-␣exp(-Sigma_ep␣/␣2)
␣␣␣␣␣␣␣␣"""
        sig = self.sigma_ep(lambda_ep, theta_D, T)
        return 1.0 - np.exp(-sig / 2.0)

    def Tc_allen_dynes(self, lambda_ep, theta_D,
                       omega_log_eV=None):
        """Allen-Dynes␣Tc␣estimate.

␣␣␣␣␣␣␣␣Parameters
␣␣␣␣␣␣␣␣----------
␣␣␣␣␣␣␣␣lambda_ep␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Electron-phonon␣coupling.
␣␣␣␣␣␣␣␣theta_D␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Debye␣temperature␣(K).
␣␣␣␣␣␣␣␣omega_log_eV␣:␣float,␣optional
␣␣␣␣␣␣␣␣␣␣␣␣Logarithmic␣phonon␣frequency␣(eV).
␣␣␣␣␣␣␣␣␣␣␣␣If␣None,␣estimated␣as␣0.7␣*␣kB␣*␣theta_D.

␣␣␣␣␣␣␣␣Returns
␣␣␣␣␣␣␣␣-------
␣␣␣␣␣␣␣␣float
␣␣␣␣␣␣␣␣␣␣␣␣Tc␣in␣Kelvin.
␣␣␣␣␣␣␣␣"""
        if omega_log_eV is None:
            omega_log_eV = 0.7 * KB * theta_D
        mu = self.mu_star
        num = -1.04 * (1 + lambda_ep)
        den = lambda_ep - mu * (1 + 0.62 * lambda_ep)
        if den <= 0:
            return 0.0
        Tc = (omega_log_eV / (1.2 * KB)) * np.exp(
            num / den)
        # Strong-coupling correction
        f1 = (1 + (lambda_ep / 2.46 /
              (1 + 3.8 * mu))**1.5)**(1/3)
        f2_arg = (1 - lambda_ep**2 * (1 + 0.2 * mu) /
                  (lambda_ep**2 + 1.44))
        f2 = 1 + f2_arg if f2_arg > 0 else 1.0
        return Tc * f1 * f2

    def evaluate(self, lambda_ep, theta_D,
                 n_channels=1, T_target=300.0,
                 omega_log_eV=None):
        """Evaluate␣R_tau␣for␣a␣material.

␣␣␣␣␣␣␣␣Parameters
␣␣␣␣␣␣␣␣----------
␣␣␣␣␣␣␣␣lambda_ep␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Electron-phonon␣coupling␣constant.
␣␣␣␣␣␣␣␣theta_D␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Debye␣temperature␣(K).
␣␣␣␣␣␣␣␣n_channels␣:␣int
␣␣␣␣␣␣␣␣␣␣␣␣Number␣of␣pairing␣channels␣(>=1).
␣␣␣␣␣␣␣␣T_target␣:␣float
␣␣␣␣␣␣␣␣␣␣␣␣Target␣temperature␣(K),␣default␣300.
␣␣␣␣␣␣␣␣omega_log_eV␣:␣float,␣optional
␣␣␣␣␣␣␣␣␣␣␣␣Log-average␣phonon␣frequency␣in␣eV.

␣␣␣␣␣␣␣␣Returns
␣␣␣␣␣␣␣␣-------
```

```python
        RtauResult
            Screening result.
        """
        Tc = self.Tc_allen_dynes(
            lambda_ep, theta_D, omega_log_eV)
        sig = self.sigma_ep(
            lambda_ep, theta_D, T_target)
        td = self.tau_dec(
            lambda_ep, theta_D, T_target)

        # Proxy formula
        Rtau_s = self.C * Tc * theta_D / T_target**2
        Rtau_m = n_channels * Rtau_s

        # Uncertainty from lambda +/- 20%
        Tc_hi = self.Tc_allen_dynes(
            1.2 * lambda_ep, theta_D, omega_log_eV)
        Tc_lo = self.Tc_allen_dynes(
            0.8 * lambda_ep, theta_D, omega_log_eV)
        Tc_unc = (Tc_hi - Tc_lo) / 2.0

        return RtauResult(
            Rtau_single=Rtau_s,
            Rtau_multi=Rtau_m,
            n_channels=n_channels,
            tau_dec=td,
            Sigma_ep=sig,
            Tc_estimated=Tc,
            Tc_uncertainty=Tc_unc)

    def screen_batch(self, materials):
        """Screen a batch of materials.

        Parameters
        ----------
        materials : list of dict
            Each dict has keys: 'name',
            'lambda_ep', 'theta_D', 'n_channels'.

        Returns
        -------
        list of (str, RtauResult)
            Sorted by Rtau_multi descending.
        """
        results = []
        for mat in materials:
            res = self.evaluate(
                lambda_ep=mat['lambda_ep'],
                theta_D=mat['theta_D'],
                n_channels=mat.get('n_channels', 1),
                T_target=mat.get('T_target', 300.0))
            results.append((mat['name'], res))
        results.sort(
            key=lambda x: -x[1].Rtau_multi)
        return results


# --- Example usage ---
if __name__ == '__main__':
    calc = RtauCalculator()

    # Known superconductors
    known = [
        {'name': 'Nb',
         'lambda_ep': 0.82, 'theta_D': 276,
         'n_channels': 1},
        {'name': 'MgB2',
         'lambda_ep': 0.87, 'theta_D': 750,
         'n_channels': 1},
        {'name': 'YBCO',
         'lambda_ep': 2.0, 'theta_D': 410,
         'n_channels': 2},
        {'name': 'H3S (155 GPa)',
         'lambda_ep': 2.19, 'theta_D': 1330,
         'n_channels': 1},
        {'name': 'LaH10 (170 GPa)',
         'lambda_ep': 2.60, 'theta_D': 1200,
         'n_channels': 1},
        {'name': 'La3Ni2O7 (14 GPa)',
         'lambda_ep': 1.2, 'theta_D': 450,
         'n_channels': 3},
        {'name': 'LaNi5H6 (prediction)',
         'lambda_ep': 0.75, 'theta_D': 264,
         'n_channels': 3},
    ]

    print("=" * 55)
    print("R_tau Screening Results (T = 300 K)")
    print("=" * 55)
    results = calc.screen_batch(known)
    for name, res in results:
        print(f"\n{name}:")
        print(f"  R_tau(single) = {res.Rtau_single:.4f}")
        print(f"  R_tau(multi)  = {res.Rtau_multi:.4f}")
        print(f"  tau_dec(300K) = {res.tau_dec:.4f}")
        print(f"  Tc_est = {res.Tc_estimated:.1f}"
              f" +/- {res.Tc_uncertainty:.1f} K")
        print(f"  Room-T SC: "
              f"{'YES' if res.Rtau_multi >= 1 else 'NO'}")
```

Listing 1. rtau_calculator.py — R_τ screening tool.

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